

Professional Self-Assessment

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Throughout my time in the Computer Science program at Southern New Hampshire University, I have developed a strong foundation in software development, problem solving, and technical communication. The courses in the program exposed me to a variety of technologies and concepts, including software engineering, databases, algorithms and data structures, security, web development, testing, and computer graphics. As I progressed through the program, I gained confidence in my ability to design, build, and improve software systems while applying industry best practices. This ePortfolio represents both the technical skills I have developed and the growth I have experienced as a software developer.

The artifact selected for this portfolio is the Travlr Getaways web application, originally created in CS-465 Full Stack Development. I chose this project because it provided opportunities to demonstrate skills across multiple areas of computer science while working with a modern full-stack application. The project uses Node.js, Express, Angular, MongoDB, and JWT authentication, making it a strong example of real-world software development. By enhancing a single artifact in the areas of software design and engineering, algorithms and data structures, and databases, I was able to demonstrate how different computer science disciplines work together to create a secure, maintainable, and scalable application.

One of the most important skills I developed throughout the program was communication. Software development is not only about writing code but also about explaining technical decisions, documenting solutions, and communicating effectively with stakeholders. Throughout my coursework, I created technical documentation, project reports, presentations, and enhancement narratives that required me to explain complex concepts in a clear and organized manner. These experiences strengthened my ability to communicate technical information to both technical and non-technical audiences.

Although many projects were completed independently, I also gained experience with tools and practices commonly used in collaborative development environments. Version control with Git, software development lifecycles, testing methodologies, and project documentation helped me understand how professional software teams develop and maintain applications. These experiences improved my ability to contribute to larger projects and prepared me for working in team-based development environments.

The software design and engineering enhancement focused on improving the architecture, maintainability, and security of the Travlr Getaways application. I implemented JWT authentication middleware, role-based authorization middleware, and standardized API response

handling. These improvements reduced duplicated code, improved maintainability, and strengthened application security by ensuring only authorized users could access protected resources. Through this enhancement, I demonstrated skills in software architecture, backend development, modular design, and secure application development.

The algorithms and data structures enhancement focused on improving how trip data is processed and presented to users. I enhanced the Angular trip listing component by implementing searching, filtering, and sorting functionality. These additions improved the user experience while demonstrating how algorithms can be applied to efficiently process collections of data. This enhancement strengthened my understanding of data manipulation, application performance, and the practical use of algorithms within a full-stack application.

The database enhancement focused on improving data integrity, validation, and performance. I added MongoDB indexes to frequently queried fields and strengthened schema validation through required fields and enumerated values. These improvements help ensure data consistency while improving query efficiency. Through this enhancement, I demonstrated skills in database design, schema development, data validation, and performance optimization.

Security has been an important theme throughout my academic experience and is reflected across all three enhancements. Authentication, authorization, validation, and structured error handling were incorporated to help protect application resources and improve overall reliability. These improvements demonstrate my understanding of secure software development practices and the importance of protecting both users and application data.

The three enhancements work together to create a stronger and more complete application. The software engineering enhancement improves maintainability and security, the algorithms and data structures enhancement improves usability and data processing, and the database enhancement improves performance and data integrity. Together, they transformed the original artifact into a more robust and scalable system while demonstrating the broad range of skills I developed throughout the Computer Science program.

As I complete my degree, my goal is to pursue a career in software development, particularly in full-stack development. I enjoy building applications that combine frontend and backend technologies to solve real-world problems and create meaningful user experiences. The projects completed throughout this program have provided valuable experience with modern development tools, frameworks, and methodologies that will help me continue growing as a developer. This ePortfolio serves as a reflection of my academic journey, my technical abilities, and my readiness to contribute to the software development profession.